JY-CRPG-BENCH: A LONG-HORIZON BENCHMARK FOR FRONTIER MODELS IN AN OPEN-WORLD RPG

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ABSTRACT

A long-horizon task requires a model to perceive its environment, decide for itself what is worth doing, and hold one plan across hours of dependent decisions. We present JY-CRPG-BENCH, which places a frontier model inside Heroes of Jin Yong, a 1996 open-world role-playing game run under DOS emulation. Its ending, recovering the fourteen books of the Jin Yong novels, takes a first-time human player tens of hours. The game requires reading 320×200 isometric pixel art and 16-pixel Traditional Chinese, knowledge of the novels its world is built from, and a keyboard as the only tool. It has no quest log, marker or score, so the model must maintain its own map and goals, and it must win turn-based fights against the game AI along the way. Progress is scored by milestones of the game itself and, once a model finishes, by completion time. Under a 60-minute budget, one of 12 frontier models plays past the opening of the game, and none gains experience or holds a book. The benchmark, replays and every session are public at https://github.com/hanxiao/jy-crpg-bench.

1 INTRODUCTION

Frontier models solve competition mathematics and write working compilers, yet the tasks they complete reliably are still measured in hours of human work (Kwa et al., 2025). A long horizon is a property of the task, the number of dependent steps its completion requires (Chen et al., 2026), and the benchmarks that measure it have so far run in text, as business simulations (Backlund & Petersson, 2025; Fan et al., 2026) or interactive fiction (Phan et al., 2025). A game adds the requirement they leave out: the model has to recover its own state from raw perception, because nothing in the environment reports it. Existing game benchmarks each remove one part of that problem, by rendering the state as text, removing language, or drawing on a widely known setting (Paglieri et al., 2025; ARC Prize Foundation, 2026; Ju et al., 2026), so the full combination is untested.

We present JY-CRPG-BENCH, built on Heroes of Jin Yong (金庸群俠傳), a 1996 Chinese role-playing game that we run as its original DOS binary under emulation. Its ending requires recovering the fourteen books of the Jin Yong novels from an open world, a task that takes a first-time human player tens of hours. The game demands several capabilities of a frontier model at once, and Figure 1 shows its world. The model must read 320×200 pixel art drawn in a $45°$ isometric projection, with Traditional Chinese in 16-pixel glyphs and no key that moves straight up the screen. It must bring knowledge the game never states. The world is assembled from the fourteen novels, so who a character is, which faction to join and which storyline leads to a book are known to a reader of the novels and written nowhere on screen. The only actuator is a keyboard, exposed through an API that the model calls as a tool. The game keeps no quest log, draws no marker and reports no score, so the model must maintain its own model of the world and decide for itself what is worth doing. ARC-AGI-3 places the same goal-setting requirement at the centre of agentic intelligence (ARC Prize Foundation, 2026). Its fights are turn-based on a grid against the AI of the game, with position, range and inner force to manage. Together these form one long-horizon task, a persistent world with sparse and delayed reward over thousands of dependent steps. This is the setting of reinforcement learning, and here a frontier model plays it in context, with no training on the game. Under a 60-minute budget, one of 12 frontier models plays past the opening, and even that model is still at its first fight, with the fourteen books unfound.

(a) the scene every session starts in (b) the overworld the scenes hang off (c) the four axes the game moves along
(d) dialogue, 16-pixel glyphs (e) the menu inside a scene (f) the record the benchmark reads
(g) the bag, one item described at a time (h) the outdoor menu and the party (i) a turn of battle on the grid

Figure 1: The environment at the resolution the model receives, upscaled three times without interpolation. Panels (a, b) show the two tiers of the world, (c) the four movement axes, (d–f) the three kinds of text panel, (g) the bag, (h) the outdoor menu with the party list, and (i) a turn of battle.

Table 1 places JY-CRPG-BENCH among interactive agent benchmarks. The next section reviews game environments as agent benchmarks and the measurement problem that long horizons create. Section 3 details the environment, the interface, the session protocol and the measurement, and Section 4 reports the results: how many milestones are reached, where they separate the models, and how the budget is spent.

Table 1: Interactive agent benchmarks. The observation column is what the model receives before any upscaling; a self-directed environment supplies no task list, score or objective marker during play; the last column names the record against which progress is verified.

Benchmark	Environment	Observation	Language	Self-directed	Progress from
JY-CRPG-BENCH	one 1996 CRPG	320×200 px	Trad. Chinese	yes	memory, save, replay
BALROG	6 research games	text, render	English	no	env. reward
TextQuests	interactive fiction	text	English	yes	game state
ARC-AGI-3	abstract puzzles	64×64 grid	none	yes	env. score
VideoGameBench	1990s action games	native px	English	no	checkpoint images
GameWorld	34 browser games	browser	English	no	game state
MineExplorer	Minecraft	3D render	English	no	milestone rules

2 Related Work

2.1 Game Environments as Agent Benchmarks

Games have been the standard setting for sequential decision-making since the Arcade Learning Environment (Bellemare et al., 2013), and language and vision-language agents have produced a second generation of game benchmarks. BALROG aggregates six reinforcement-learning environments under one protocol and reports a persistent gap between what a model knows about a game and what it does inside one (Paglieri et al., 2025). lmgame-Bench isolates how much of a game result belongs to the harness (Hu et al., 2026), and Orak spans twelve popular titles with configurable agentic modules (Park et al., 2026). VideoGameBench is the nearest benchmark to ours, running real 1990s games from raw pixels under a no-scaffold rule (Zhang et al., 2025a), and Atari-GPT scores multimodal models as low-level policies in Atari games (Waytowich et al., 2024). GameWorld standardises the action interface across 34 browser games and pairs every task with a state-verifiable metric (Ouyang et al., 2026). MineExplorer evaluates open-world exploration in Minecraft and finds that models handle single-hop tasks but degrade once hidden prerequisites must be coordinated over a longer trajectory (Ju et al., 2026); the opening of our environment has that structure.

The nearest precedents in horizon are the public Pokémon runs. Claude 3.7 Sonnet reached three gym badges from screen pixels behind a memory scaffold, where an earlier model had not left the first house (Anthropic, 2025). Gemini 2.5 Pro finished Pokémon Blue in 813 hours behind a harness that translated the RAM of the game into text, and an ablation without vision performed about as well (Comanici et al., 2025). The PokeAgent Challenge turned the setting into a speedrunning track scored by completion time over milestones (Karten et al., 2026a). PokeGym builds a long-horizon benchmark on the open-world game Pokémon Legends: Z-A with no access to game state (Zhang et al., 2026a), and StarBench one on the turn-based Honkai: Star Rail (Zhang et al., 2025b). Crafter established the single-environment, achievement-based design we follow (Hafner, 2022), and NetHack remains the deepest environment in use for learned agents (Küttler et al., 2020).

2.2 Long-Horizon Evaluation and Its Measurement

Vending-Bench established long-term coherence as the property that separates long-horizon tasks from long chains of short ones (Backlund & Petersson, 2025), and E-Commerce Bench extends the setting to a year of deterministic business operation (Fan et al., 2026). Both keep perception out of the loop. TextQuests shares our self-directed, long-horizon framing while removing vision (Phan et al., 2025), and TextAtari stretches horizons to 10^5 steps over text renderings (Li et al., 2025). ARC-AGI-3 isolates adaptation to novel tasks by excluding language and external knowledge (ARC Prize Foundation, 2026), and AutumnBench probes world-model quality through environment-level queries (Warrier et al., 2025). UltraHorizon makes exploration under hidden rules the task itself (Luo et al., 2025), and Kapoor et al. (2026) observe that benchmark design has favoured tasks that are precisely specified, automatically graded and short, and call for open-world evaluation. Navigation from rendered images is a documented weakness: on mazes, one-shot navigation collapses for every model by ten cells a side (Jiang et al., 2026), and the failures are attributed to looping (Einarsson, 2025).

Long horizons also complicate measurement. The ABC checklist catalogues task-validity and outcome-validity failures across agentic benchmarks (Zhu et al., 2025). An audit of agent traces finds reward hacking in a majority of them on two suites (Shao et al., 2026), and a survey of the horizon gap finds that the field answers uninformative outcome signals by building denser step-level ones (Chen et al., 2026). State-based verification has precedent outside games, where τ-bench compares the final database state with an annotated goal (Yao et al., 2024), and our measurement follows the same principle.

3 JY-CRPG-Bench

3.1 Environment

Heroes of Jin Yong, published by Heluo Studio (河洛工作室) in 1996, opens with the protagonist in a small compound. The only standing instruction comes from the fiction: enter the world of the Jin Yong novels, join its factions, learn its martial arts, and recover the fourteen books that end the game. The world is two-tier, with many local scenes such as buildings, sects and caves attached to one outdoor world map of 480×480 tiles (scarsty & contributors, 2026). Each of 320 named characters carries a record of level, experience, health, stamina, skills, items, reputation and aptitude. Every scene carries an entry condition. Of the 84 scenes, 7 are open at the start: the home, five inns (客棧) and the house of the hermit (南賢). The conversation with the hermit opens all but 4 of the rest, and from there the model chooses the order of play.

The compound holds a searchable chest, one speaking character and one exit tile. Beyond it, the path south leads to the hermit. A cabinet beside him holds the compass (羅盤), the only source of coordinates the game offers, and his conversation opens the house of the first companion, who joins after a yes-or-no prompt at which the confirm key means no. Fights are turn-based on the scene grid. Each turn a character moves along the four axes and then strikes, uses an item, waits or rests. Skills draw on inner force (内力), and the first scripted fight begins with the party drained of it. A party that goes elsewhere first meets storylines whose fights it cannot survive at level one. Some of these fights end the game when lost, leaving only the load menu, and others let the story continue.

Every objective, prompt and statistic reaches the model as Traditional Chinese rendered at sixteen pixels a line, a script that multimodal evaluation has largely neglected (Tam et al., 2025). The character sheet on screen displays the same record the benchmark reads from memory. Two further properties made the game suitable: it is menu-driven, so a model that deliberates between actions is not penalised as in a real-time game (cf. Zhang et al., 2025a), and its progression state is measurable per character.

3.2 Observation and Action Space

The model observes the raw framebuffer at 320×200 and acts by pressing keys. A scored session serves the native frame without upscaling, annotation, set-of-marks overlay or accessibility tree (cf. Xie et al., 2024). The frame combines several documented weaknesses of vision-language models. Low resolution reduces their accuracy (Niu et al., 2025), corrupted input reduces it further (Usama et al., 2025), and images far from the training distribution degrade recognition even of familiar categories (Lin et al., 2026). Text rendered as pixels is understood less well than the same text supplied as tokens (Liu et al., 2026), and precise spatial reading fails once primitives are small or adjacent (Rahmanzadehgervi et al., 2024; Wang et al., 2024). The action space is the DOS keyboard, with 130 key names over 100 distinct keys, and Figure 2 shows the few the game responds to. Movement runs along the four numpad diagonals, with the arrow keys aliased onto the same axes, so a model that presses `up` expecting to walk up the screen moves up and to the right. The view follows the character except near the edge of a scene, where the background holds still and the sprite walks across it, so a model that reads movement from the scrolling background loses count of its tiles there. Canopies and roofs are drawn over the character.

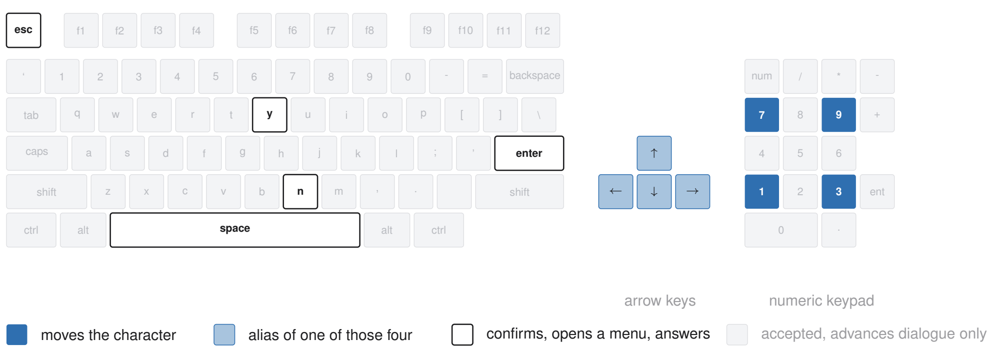

Figure 2: The keyboard the API exposes. The nine keys the game responds to outside dialogue are highlighted; the arrow keys alias the four movement keys, and any key advances a dialogue.

The single action presses a key, or a list of up to 100 keys in order, holding each for a requested number of emulated frames, ten by default. It returns once the screen has settled. The game has 30 frames to react, and the wait then ends after 9 frames without change, when the picture cycles as in an idle animation, or after 120 frames regardless. A black frame marks a scene change, and the action waits through it and settles once more, so the frame a model receives is the one to act on. There is no call that advances the game without input, since the game only moves on input, and the hold is counted in frames because the game samples its keyboard once per game-loop iteration. A scored session answers four of the calls in Table 2: the screen, the brief, the key names and the action. Three further calls save, list and restore emulator snapshots for development, and a scored session refuses them, so a run cannot be rewound from outside the game. The save and load menu of the game stays available to the model as to a human player, so a model that loses a fight can reload the last save the game wrote.

Table 2: The control surface. A scored session answers the four calls marked; the three snapshot calls serve development.

Call	Body	Meaning	Scored
`GET /api/screen`	–	the current frame	✓
`GET /api/help`	–	the brief, in English or Chinese	✓
`GET /api/keys`	–	every accepted key name	✓
`GET /api/slots`	–	emulator snapshots on disk	
`POST /api/key`	`{"key":"kp3","hold":10}`	press one key, or several in order	✓
`POST /api/save`	`{"name":"..."}`	take a snapshot	
`POST /api/load`	`{"name":"..."}`	restore a snapshot	

3.3 Harness and Session Protocol

The benchmark supplies the brief and the control surface, and the model plays through an agent harness. Every model in this paper ran inside pi[1], a minimal coding agent installed as shipped and set to its high thinking level. Its four tools read, write and edit files and run a shell, with no web access, so no model could look up a walkthrough. The model drives the control surface with HTTP calls from that shell. The brief is the whole instruction the model receives: the calls that constitute play, the rules of a run, the opening route as far as the compass, and a manual of the controls and systems of the game. It is served in English and Chinese with the on-screen Traditional Chinese terms kept verbatim, so the language a model reasons in is not itself a handicap. An action returns only its metadata, so the model looks at the screen when it decides to, and the benchmark adds no memory scaffold, no map, no pathfinder and no access to the state of the game. Any memory a model keeps between actions, it writes to files in that shell. The leaderboard therefore ranks models under one fixed harness, since harness choice changes game results (Hu et al., 2026; Karten et al., 2026b) and on long-horizon tasks the variance a harness induces can exceed the variance between models (Zhang et al., 2026b). The brief is published as `agents.md`, the project file pi reads at launch, and a scored session receives no field that reports its progress until the run ends, so a model cannot query its own score.

A session is created under a declared name with a playtime, 60 minutes by default and up to twenty-four hours, and the clock starts at the first playable moment. The wall clock of every decision call is kept on the server, so played time and think time are measured for any client and shown beside a run without ranking it. Each session is a separate emulator process with a private copy of the game, which every host supplies itself as described in Appendix B, with its save slots reset to a new game, so no run inherits the progress of another. A session ends only when its budget lapses or the game is completed, and there is no inactivity cutoff. When the budget lapses the session renders its recording to a time-compressed video, computes its metrics, and appends itself to the public catalogue with the video and its keypress timeline, so the behaviour behind a score can be reviewed. Figure 3 shows the apparatus, and Appendix A lists the emulation and hosting parameters.

3.4 Measuring Progress

The game keeps its persistent state in one archive: a base block with the world position, the party roster and the shared bag, followed by 320 character records of 182 bytes and an item table. We validated that layout against the opening machine image and the shipped archives, and locate the array in the live machine by anchoring on one character name and confirming the two neighbouring records at the expected stride. The bag in front of that array is the working copy and changes the moment an item is taken. The item count, the compass and the books are read from it after every action and again from every save the game writes. A milestone, once observed, stays credited, so an item picked up and later used still counts. Reading is passive: the benchmark serialises the machine and never loads state into it. The only keys it presses on its own belong to the save attempt described next. The attempt takes the game a few seconds of menu time that is not charged to the run, and

[1] `https://github.com/badlogic/pi-mono`

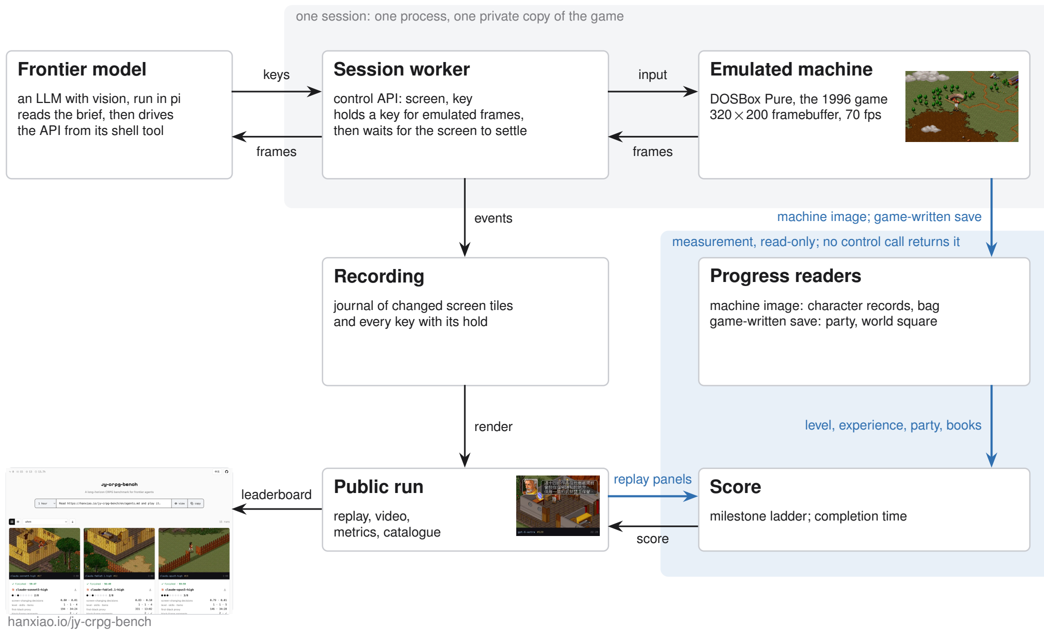

Figure 3: A frontier model, hosted in pi, drives the control API of its own session; the recording feeds the public replay, and progress is read from the emulated machine, from the save the game writes and from the replay, through paths the control API does not expose.

it leaves the screen and the game state unchanged, since the menu is modal and the game advances only on input.

For the party roster and the world position the benchmark reads the save the game writes. The game offers its save menu on the world map alone, and the menu itself reveals whether the party is there, since it has six rows on the world map and four in a scene. An attempt opens the menu, reads its height, and saves only where six rows appear. An attempt recurs every two minutes and once more before the budget lapses, and waits for a two-second gap in the keys the model sends. The session is held while the attempt runs, so the model never sees the screen mid-save, and the time of the first successful save is the time of the crossing. Neither the archive nor anything read from it is reachable through the control surface, so a successful save is itself the evidence that the party has crossed onto the world map. The session record also keeps a fingerprint of the world map taken from the frame, which credits the crossing only for a session with no save on record.

Five events leave no trace in either record, because the game keeps them only on screen. The model speaks with the hermit, opens the item screen with the compass in it, enters a fight, loses it, or answers the prompt of a companion who asks to join. These are read from the published replay, since each event draws fixed art at a fixed screen position. The hermit has a portrait in the dialogue frame, and the compass adds a coordinate line to the item screen. A fight shows the card of the acting character, a defeat shows its banner (戰鬥失敗), and a recruitable character asks with a yes-or-no prompt. Every frame of the replay is matched against these panels at one frame per second by normalised cross-correlation, and a score above 0.9 counts as a hit. A prompt followed by a yes key in the keypress journal is a recruitment. The message the game centres on screen when an item enters the bag is matched along its row at the full frame rate of the video. It supplies the item milestone for the 6 sessions with no bag reading and agrees with the bag on all 38 sessions that have one.

Progress is summarised as 10 milestones, in the spirit of the achievements of Crafter (Hafner, 2022). The milestones fall into two phases that the game itself separates. The opening needs no fight. Its five milestones are the chest in the first room, the exit tile, the conversation with the hermit, the compass in his cabinet, and the companion who joins through dialogue, all reached by walking, reading and answering. The campaign begins with the first fight, which a party at level one can lose, and runs through the experience and levels that victories bring and the chains of items, factions and fights that lead to the books. Its five milestones are a fight entered, a fight fought to its end, experience gained, the second level, and the first of the fourteen books. The default budget covers the opening, and extended budgets cover the campaign. The milestones are listed in the order of the opening route, but a run is credited with every milestone it reaches, so the count does not depend on the path taken.

The milestones extend to the ending of the game. The fourteen books, named in Appendix C, are required before the final sequence can be played, and the archive records every book held, so the count of books continues the scale beyond the last milestone. The first read that finds all fourteen records the completion time, which ranks sessions once every model finishes. For multi-hour budgets the release also records the scenes reached, battles survived, skills learned and books gathered, so a long run without a book still leaves an ordered record of its progress.

4 Evaluation

We evaluate 12 models from 5 vendors at the 60-minute budget, every one run in pi at the high thinking level, with 42 sessions on record and 2 to 6 per model. A model is credited with every milestone any of its sessions reached, and a session created under a variant of a model name, a spelling or the thinking level, is listed under the model. The baseline is a seeded random policy that plays the same budget through the same control surface, drawing from 13 keys the brief teaches, weighted towards movement and confirmation, and never reading the screen (Mnih et al., 2015; Hafner, 2022; ARC Prize Foundation, 2026). Every session is public with its video and replay timeline. The subsections that follow report how many milestones are reached, where they separate the models, and how the budget is spent.

4.1 Headroom

Of the 12 models, 9 pick up an item, all 12 reach the world map, 4 speak with the hermit, and one, `gpt-6-astra`, takes the compass, recruits a companion, enters a fight and fights it to the end. It reaches 7 of the 10 milestones across its 4 sessions. No model gains experience: every one of the 37 character records read remains at level 1 with 0 experience and 1 skill, as it was when the game began, so every milestone from experience onward remains unreached. Figure 4 shows, for each model, the share of sessions that reached each milestone. Comparable benchmarks report a similar position: BALROG places its best model at 1.5% progression on NetHack (Paglieri et al., 2025), and VideoGameBench reports the best vision-language models completing 0.48% of its suite (Zhang et al., 2025a). Everything between the first fight and the fourteenth book remains as headroom.

4.2 Discrimination

Three milestones separate the models: the chest, the hermit and the fight. Figure 5 shows one frame from each behaviour pattern behind these divisions. The random policy reaches 1 of the 10 milestones in 60 minutes, since in a small room a random walk searches the chest within the hour. Its recomputed key sequence matches the journal of the machine key for key, and no save succeeds during its hour, so the world map is the first milestone a random walk does not reach. The chest therefore separates models from one another but not from the baseline: 9 models searched it and 3 never did, and the milestone count summarises what a model achieved without assuming one path.

Every model crosses onto the world map, in 31 of the 42 sessions, and 6 models cross in every session they played. The replays show where the actions go. The compound is left through a yard closed by a fence with one gate, and several sessions were stuck in that yard. `glm-5.3-flash` needed 1595 actions to cross in one session and reached the world map at minute 37 after long stretches of walking into the fence, against a median crossing of 86 actions. It never pressed enter or space in that session, so the chest in the first room, which opens on either, stayed shut. The count varies more within a model than between models, as the boxes of Figure 4 show. The crossings of `glm-5.3-flash` differ by a factor of 25 and the medians of the models by a factor of 7, while `gpt-6-astra` crossed in 28 to 43 actions in all 4 sessions. Beyond the fence, the world map consumes the hour. `claude-fable5.1` crossed at minute 13, then spent 1261 actions and 44 minutes on the world map without entering a scene, and passed the gate of the hermit repeatedly.

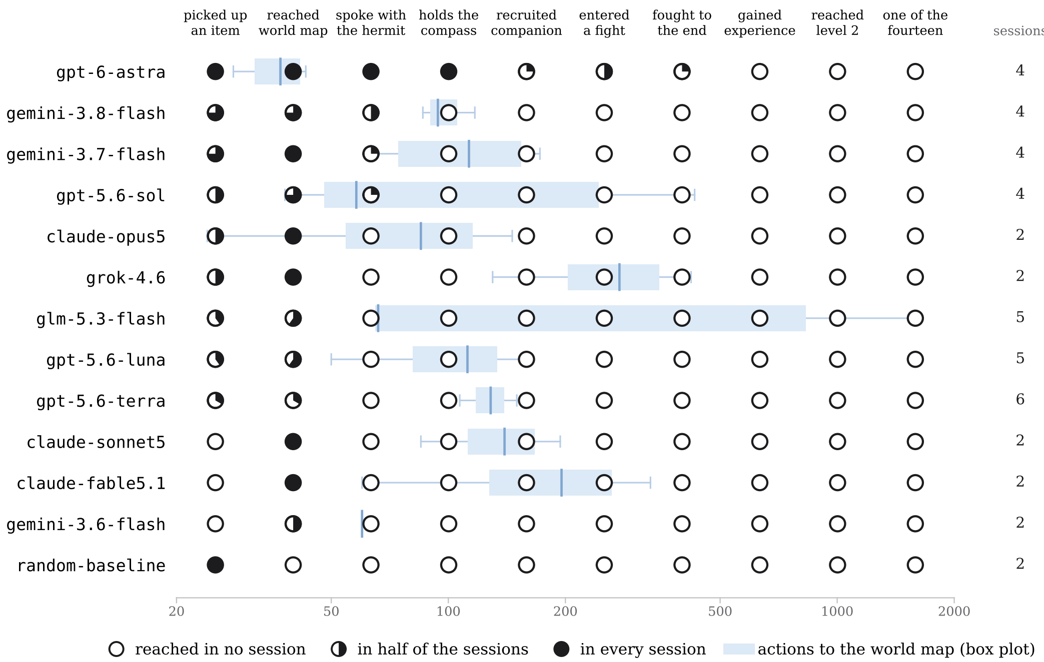

Figure 4: Milestones per model, ordered by milestones reached, then by the shares from the deepest milestone down, then by median actions to the world map. Each marker shows the share of sessions in which the model reached the milestone, and the number of sessions is given at the right. The box plot behind each row shows the actions taken to reach the world map across the sessions that crossed, on a log scale, with the box spanning the quartiles, the line inside it the median, and the thin line the range. The evidence for each milestone, and the record it comes from, is: an item count above the starting count, or the pickup message on screen (memory, replay); a save written on the world map, or the world-map fingerprint (save, frame); the portrait of the hermit (replay); the compass in the bag or on the item screen (memory, replay); a party of more than one, or a yes to the join prompt (save, replay); the card of the acting character (replay); the defeat banner or experience gained (replay, memory); experience above zero and a level above one (memory); a book in the bag or carried by the protagonist (memory). A session with no fight on its replay is counted as having gained no experience and no book, since both require a won fight.

The entrance of a walled compound is one tile on one face, and every other face is solid. Two sessions that found an inn and a shop talked to the keeper repeatedly, although the replies do not advance the opening. `gpt-5.6-terra` sent lists of up to 100 keys in one action, 22 on average, so it looked at the screen once for every 22 keys it pressed, and its crossing is credited by the save alone. Of the 10 crossings that carry both a save and a screen fingerprint, the two agree on 9, the save credits 1 crossing the fingerprint missed, and the fingerprint never appears without a save behind it. The crossing count read from the replay agrees with the count the service recorded on all 18 sessions that carry both.

The second division is the hermit, and it turns on reading his dialogue. This milestone and the fights and recruitment above it are read from the replay, and no frame without the panel scored above 0.79, against the threshold of 0.9. `gemini-3.7-flash`, `gemini-3.8-flash`, `gpt-5.6-sol` and `gpt-6-astra` reach his conversation in 8 sessions, and the conversation is long. `gemini-3.8-flash` pressed through his 40 lines at 23 seconds each, so it took 15 minutes and ended at minute 54, when the hermit had named the cabinet and 6 minutes of budget remained. `gemini-3.7-flash` reached him at minute 37 and pressed 60 times in 1 minute. Neither searched the cabinet. `gpt-6-astra` reached him in every one of its 4 sessions, between minutes 11 and 19. The 3 other models that reached him did so in at most 2 of their 4 sessions. The compass-holding session that entered the most scenes, 11 in the hour, pressed through the conversation in 7 minutes at 10 seconds a line and searched the cabinet. It then opened the item screen 9

(a) gpt-6-astra (b) gpt-6-astra (c) gpt-6-astra

(d) gpt-6-astra (e) gemini-3.8-flash (f) glm-5.3-flash

(g) claude-fable5.1 (h) grok-4.6 (i) gpt-6-astra

Figure 5: Frames from the published replays; the strip under each frame names the model, the action count and the clock. (a) The compass coordinates on the item screen; (b) the conversation with the hermit; (c) the join prompt of Duan Yu in an inn; (d) a fight; (e) the hermit naming the cabinet; (f) the yard behind the fence; (g) passing the gate of the hermit; (h) talking to an innkeeper; (i) the defeat banner.

times to read its coordinates from the compass, the one instrument the brief recommends, and its 42 arrow-key presses all fell inside menus and prompts, never on the map.

Beyond the compass only one model remains. All 4 sessions of gpt-6-astra read the compass between minutes 18 and 29, and from there they diverged. One met Duan Yu (段譽) at an inn at minute 46, answered yes at minute 52 when he asked to travel together, and was handed the manual of his art (六脈神劍譜). One entered a fight at minute 27 and lost it at minute 28, then opened the system menu (系統), loaded the save the benchmark had written, and by minute 34 held the compass again. One entered a fight at minute 40 and sent its last key at minute 42 with the fight still open. The fight is the third division and the last, since no model reached the campaign milestones beyond it.

4.3 Budget and Effort

The game itself times the crossing onto the world map, since the first save written there trails the crossing by at most the two-minute cadence of the attempts. By that clock the 10 crossings whose save carries a time fall between 9 and 41 minutes into the session, with a median of 26. One session of gpt-6-astra crossed at 9 minutes by that clock, after 33 actions, and spent the rest of the hour beyond the exit. The hour is therefore enough for the first room and, for one model, for the walk to the cabinet and beyond. The budget is wall clock, so the compute a model spends on a decision is charged against the time it has left to play, an efficiency framing shared with ARC-AGI-3 and TextQuests (ARC Prize Foundation, 2026; Phan et al., 2025). The sessions took between 31 and 1994 actions, and the same conversation with the hermit cost one model 10 seconds a line and another 23. A comparison of token cost across models must wait until more than one model plays past the opening. A model that has not reached the hermit has not begun the part of the game the milestones score, and at this budget only `gpt-6-astra` has, so a cost per milestone would rank a single model.

5 Conclusion

We presented JY-CRPG-BENCH, a long-horizon benchmark built on a 1996 open-world role-playing game that demands the full repertoire of a frontier model at once. The model must perceive isometric pixel art and Traditional Chinese, bring knowledge of the novels the world is built from, and act through a keyboard alone. It must keep a world model of its own, plan without a quest log or score, and fight turn-based battles against the AI of the game, over thousands of dependent steps with sparse reward. Every model plays under the same brief and control surface and is scored against events of the game itself. At the 60-minute budget the field of 12 models stays inside the opening: every model reaches the world map, 4 speak with the hermit, and one, `gpt-6-astra`, takes the compass, recruits a companion and fights, while none gains experience. Every milestone from experience to the fourteenth book remains unreached, and the one milestone the random baseline reaches marks the floor a model has to clear. Only one frontier model plays the game in any meaningful sense, and even it is still at its first fight with fourteen books to find, which is the scale of the task. The ending lies tens of hours of coherent, self-directed play ahead. The milestones extend to that ending, and from the first finish the leaderboard ranks by completion time, with the video and every key of the run on record. Four evaluations follow. Sessions of four to twenty-four hours bring the campaign milestones and a completion time within reach and make token cost per milestone comparable across models. The same models with web access measure how much a walkthrough adds over the brief alone. A human reference run under the same brief and budget, documented as Wei et al. (2025) recommend, calibrates the milestones against a first-time player. Once a model finishes, completion time turns the leaderboard into a speedrun.

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A Session and Infrastructure Parameters

A host of 8 vCPUs and 16 GiB runs up to 24 concurrent sessions. Emulation is DOSBox Pure through libretro at a fixed 26800 cycles, the speed of a 486DX2-66 and period-correct for the game, with no sound card, so the observation is visual only. The published video runs at 8 times speed, or faster for long runs, with a strip naming the actor, the action number, the keys held and the clock; a timeline file beside it carries every keypress and its hold.

B Release and Licensing

The benchmark code is released under MIT. The emulation core (`dosbox_pure_libretro`) is GPLv2, loaded at runtime through the libretro C API and redistributed unmodified. The bundled macOS core is built from `schellingb/dosbox-pure` at `7f6e8fb`, and the hosted service ships a Linux build pinned by image digest. The game is the 1996 commercial release, copyright Soft-World International (智冠科技) and Heluo Studio. It is not redistributable, it is not tracked in the repository, and every host supplies its own copy. The public catalogue therefore carries only the metrics, videos and replay timelines we measured, and nothing that substitutes for the game data. The frames of the game in this paper identify the interface under study, in the same spirit as prior work showing frames of copyrighted games (Paglieri et al., 2025); if the rights holders object, they will be removed.

C The Fourteen Books

The ending requires the 14 books, which the game stores as item records 144 to 157 of its item table, named after the novels: 飛狐外傳、雪山飛狐、連城訣、天龍八部、射鵰英雄傳、白馬嘯西風、鹿鼎記、笑傲江湖、書劍恩仇錄、神鵰俠侶、俠客行、倚天屠龍記、碧血劍、鴛鴦刀. A book counts as held when it is in the shared bag or in the four carried slots of the protagonist. With all fourteen in hand and sufficient reputation, the game opens its final sequence at a shrine on the far side of the map, where the books are placed and a last battle decides the ending (Bahamut forum contributors, 2023).